Science Olympiad

Remote Sensing - Division C

2025 Hawk-Hornet Invitational

Team Number __________

School____

Student Names____

Instructions:

You have 50 minutes to complete this exam.

All problems are 1 point each unless otherwise noted.

You may separate the pages of this exam, but please assemble the pages back in the proper order and re-staple when you hand them in (2-point penalty for out of order / missing pages). For the calculations, show your work and use proper units / significant Figures to receive full credit. Tiebreakers (**TB**) are noted. If it goes beyond 5^{th} tiebreak problem, the problems in reverse order will be used. Photo is the University of Michigan 1^{st} Generation Doppler Lidar (Todd Irgang, 1992).

Section 1: Remote Sensing Instrumentation and Physics (25 points)

1. (5 points) Label each of the following with A for Active sensor or P for Passive Sensor
   a. Radar ______
   b. Radiometer ______
   c. Lidar ______
   d. Visible camera ______
   e. Infrared camera ______

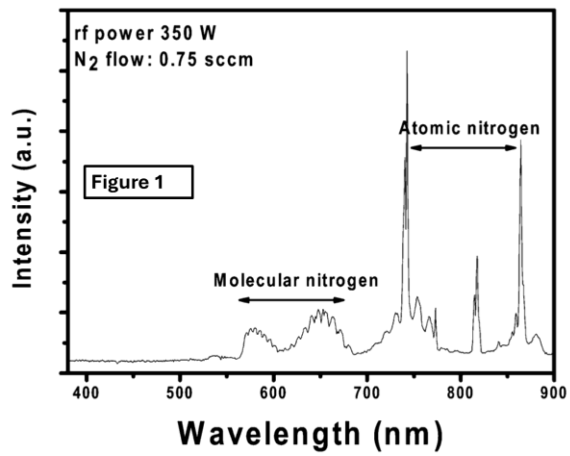
2. What type of scattering occurs when the wavelength of the emitted radiation is smaller than the size of the particle being detected?
   a. Rayleigh Scattering
   b. Mie Scattering
   c. Brillouin Scattering
   d. Raman Scattering

3. What type of scattering occurs when the wavelength of the emitted radiation is the same size or larger than the size of the particle being detected?
   a. Rayleigh Scattering
   b. Mie Scattering
   c. Brillouin Scattering
   d. Raman Scattering

4. If a visible (digital) camera were used to image a blackbody source, how much brighter would the image be if the temperature of the blackbody were tripled?
   a. 3X as bright
   b. 9X as bright
   c. 1.5X as bright
   d. 81X as bright

5. If a visible camera were used to image a blackbody source, how much would the peak wavelength shift if the temperature of the blackbody were tripled?
   a. The peak wavelength would be 3X higher
   b. The peak wavelength would be 3X lower
   c. The peak wavelength would be 9X higher
   d. The peak wavelength would be 9X lower
   e. Not enough information given to know

6. If you have a satellite equipped with radiometers with color filters to look at bands throughout the visible and near infrared portions of the spectrum, what type of orbit would be best for your instrumentation to collect the maximum amount of useful data?
    a. Geosynchronous
    b. Geostationary
    c. Sun-synchronous
    d. Polar



Refer to Figure 1 above for the next 2 problems.

7. What is displayed on this graph?
    a. Emission spectra
    b. Absorption spectra
    c. Blackbody spectra
    d. Continuous spectra

8. (2 points) What regions of the electromagnetic spectra are shown in this graph? Be specific (i.e. use two words for the higher wavelength portion).

_________ and ________________

9. (4 points – **TB1**) A polar orbiting satellite is located at an altitude of 300 miles (483 km) above the surface of the Earth. What is its orbital velocity?

10. (3 points - **TB3**) How many minutes does it take the satellite from Problem 9 to complete one orbit?

11. If the satellite drops to a lower orbit due to atmospheric drag (friction), does the velocity increase or decrease? (Circle the correct answer for 1 point)

A satellite with a polar orbit 100 miles (160 km) above the surface of the Earth has a MWIR (3 – 5 μm) imaging system with a one megapixel (1024 × 1024) format, a pixel pitch of 50 μm, and a 100 mm diameter lens with a focal length of 1250 mm. The system is diffraction-limited (i.e. the resolution limit is represented by ${}^{\lambda}/_{D}$).

12. (4 points – **TB2**) What is the ground sampling distance that results from the instantaneous field-of-view of this system?

13. What is the ground sampling distance that results from the field-of-view of this system (again expressed as an area element on the ground)?

Section 2: Image Interpretation (35 points)

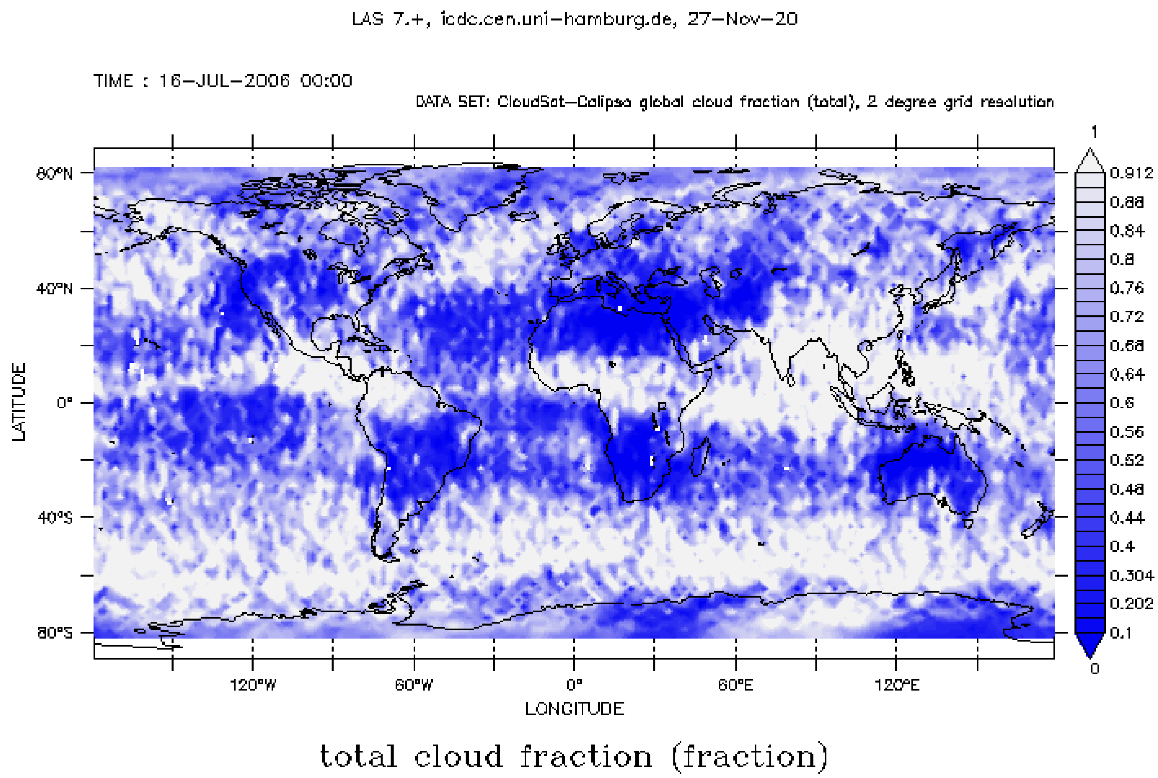


Figure 2: Sample map of the global total cloud cover distribution as obtained with CALIPSO / CloudSat for July 2006.

The above Figure combines CloudSat Radar data and Calipso lidar data. CloudSat provides a measurement of refractive index (radar cross section), which is a measure of cloud and precipitation particles at a range resolution of 240 m. Calipso-CALIOP provides 3 vertical profiles of laser energy scattered by aerosol and cloud particles at a resolution of about 15 m.

14. Are these considered active or passive sensors, or a combination of the two? (Circle the correct answer)

15. (2 points) With what (time) sampling rate must the detector on Cloudsat use? The Cloudsat radar looks straight down to achieve the 240 m resolution.

16. (2 points) What is the sampling rate of the Calipso lidar? It does not look straight down, but the 3-degree viewing angle is close enough for the purposes of this calculation.

17. (2 points) What is one benefit and one drawback of using the Cloudsat radar?

18. (2 points) What is one benefit and one drawback of using the Calipso lidar?

19. What do the blue portions of the image represent?

20. Not explicitly given, what period of time would be reasonably represented by this data set?
    a. Minutes
    b. Hours
    c. A Month
    d. A year

Answer Problems 21 – 27 based on Figure 3 below, which shows the sea surface temperatures recorded over a 4-day period by the MODIS (on Aqua) instrument during late August 2006.

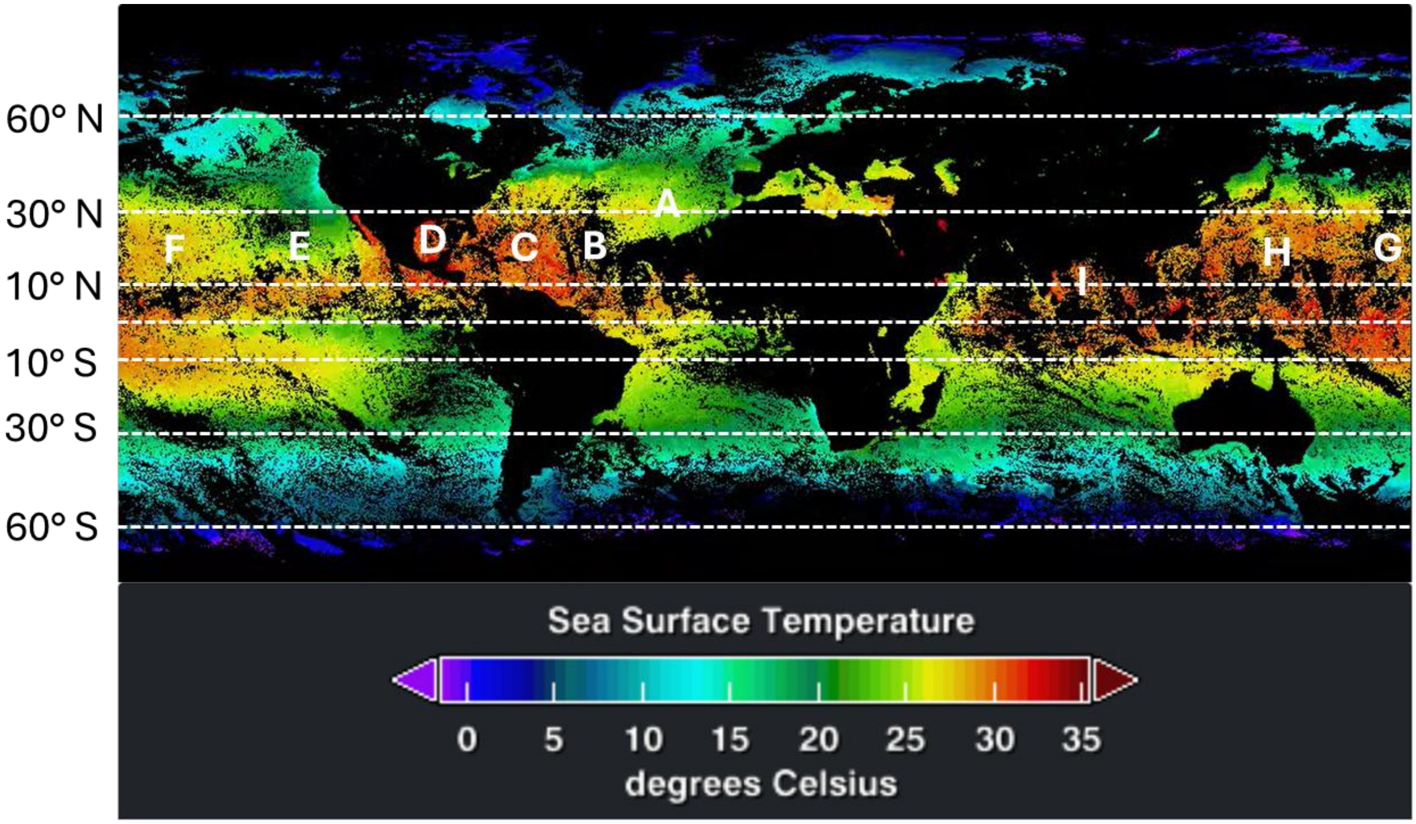


Figure 3: MODIS (Aqua) Sea Surface Temperatures from August 2006

21. (3 points) Which area(s), denoted by the letters A-I, are areas where hurricane formation could take place?

22. What is the cause of the big black spot denoted under letter B?

23. (2 points) Could hurricanes form in the large red spot south of the equator from letter G? Why or why not?

24. What is the sea surface temperature just off the Yucatan peninsula (just south of letter D)**?**

25. There is data above 60° N, but not below 60° S. What is the reason for this discrepancy?

26. (2 points) Imagine that we flip the temperatures about the equator (i.e. the areas that are red at 30° N would be red at 30° S) and advance the collection time 6 months to the Southern Hemisphere summer. If the letters were in the same spots in the South as they are in the North, would you put the same letters for this problem as you did for Problem 21? Why or why not?

27. What do they call a storm with 165 mph winds in the South Pacific?

The Advanced Technology Microwave Sounder (ATMS instrument) aboard the Suomi NPP satellite is a total power radiometer that scans 22 channels in the microwave region of the spectrum. Channel 1-16 are primarily for atmospheric temperature profiling (from the surface to 0.1 mb), while channels 17-22 primarily profile water vapor in the troposphere. Use the following set of Figures which show data taken with the ATMS instrument (Figures 4(b) and (c)), as well as supporting data (Figure 4(a)) to help you to answer Problems 28 – 38.

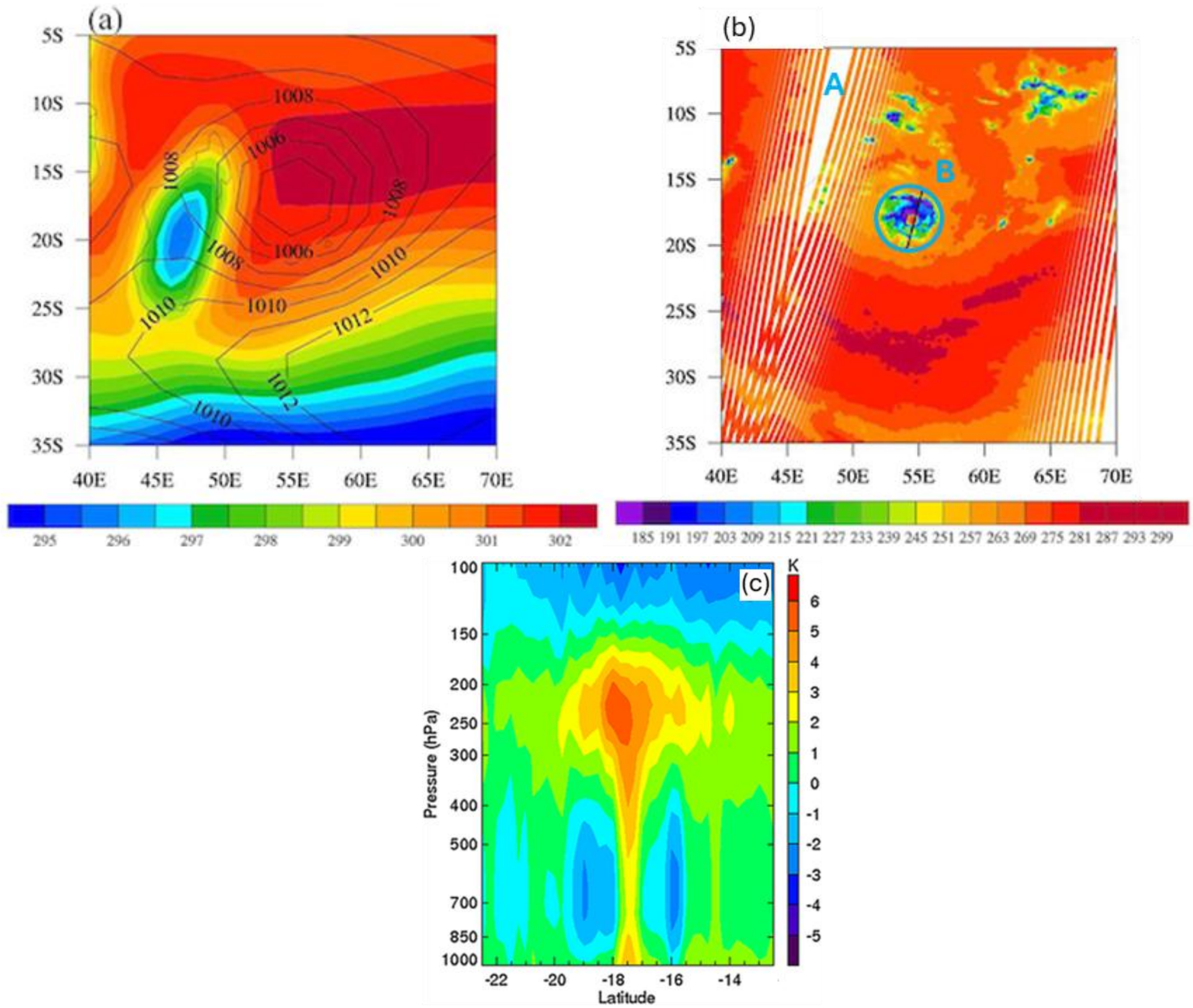


Figure 4

28. Is ATMS an active or passive instrument? (Circle the correct answer)

29. (2 points) On December 12, 2012, Tropical Cyclone Giovanna reached Category 3 on the Saffir-Simpson scale. Figure 4(a) shows the surface temperatures and pressures estimated by the National Centers for Environmental Prediction Final global analyses for the region of this storm. These estimates are an assimilation of data from many sources on a 1° × 1° (lat / long) grid**.** Which of the quantities (temperature or pressure) in Figure 4(a) is off from what you would expect, and why?

30. What is causing the region of low temperature shown in Figure 4(a)? Hint: see Figure 2.

31. Figure 4(b) shows data from ATMS channel 18. What is this plot (primarily) showing?

32. What is represented by the area marked by the letter A in Figure 4(b)?

33. Note that the very center of the region near B (denoted by the blue circle in Figure 4(b)) is red, as is most of the region surrounding B. However, there is a region of much lower antenna temperature within region B and some nearby areas. What is the reason for these lower temperatures?

34. Why is it red in the center of this region?

35. Figure 4(c) shows “temperature anomalies” – the difference between the temperature within Giovanna compared to the environmental temperature (averaged temperature within 15° latitude / longitude about the storm but without temperatures around the core region being included) measured with ATMS. At what pressure is the greatest deviation found?

36. Approximately what altitude (in km) does this correspond to?

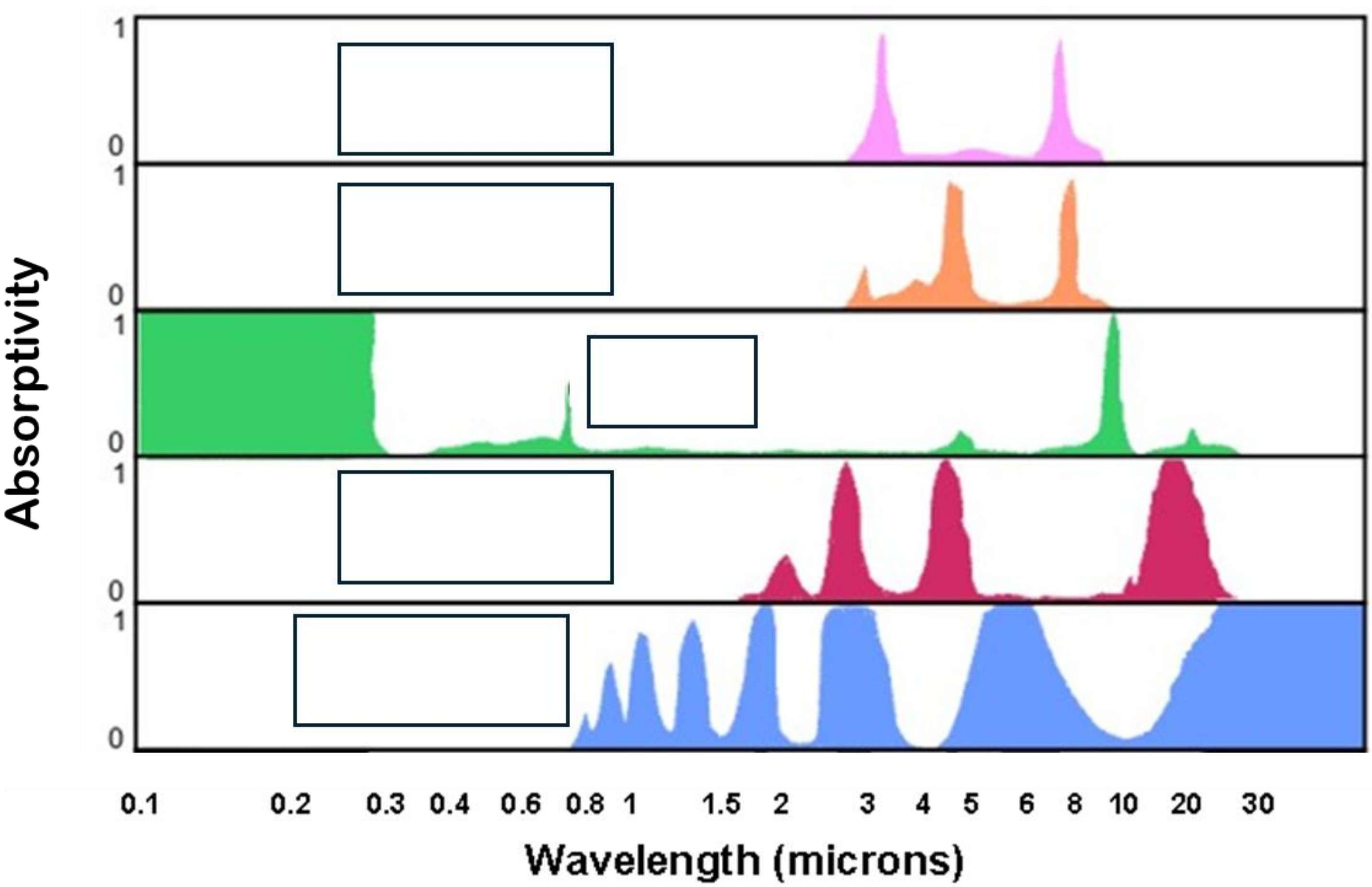
37. In their paper, Zhu and Wang comment that the region of heaviest rain and maximum sustained winds occurs where the coldest temperature anomalies are present. At what latitudes would you find these cold temperature anomalies?

38. (2 points) Based on your answer to Problem 37, what is the radius (in km) of maximum sustained winds, as measured from the hurricane center?

Section 3: Climate Processes (20 points)

39. Which of the following is true about El Nino (seasons relative to Northern Hemisphere seasons; choose 1 where all statements are true):
    a. El Nino peaks in the summertime, it is characterized by warm waters off the coast of Mexico and cold waters on the other side of the Pacific; it causes colder, wetter winters here in Michigan
    b. El Nino peaks in the wintertime, it is characterized by cold waters off the coast of Mexico and warm waters on the other side of the Pacific; it causes warmer, drier winters here in Michigan
    c. El Nino peaks in the wintertime, it is characterized by warm waters off the coast of Mexico and cold waters on the other side of the Pacific; it causes warmer, drier winters here in Michigan
    d. El Nino peaks in the summertime, it is characterized by cold waters off the coast of Mexico and warm waters on the other side of the Pacific; it causes warmer, drier winters here in Michigan

40. (5 points) The plot below shows 5 greenhouse gasses. Label them in the boxes provided.



41. Looking at their spectra, what makes them a greenhouse gas?

42. According to NASA, which gas is considered the most abundant greenhouse gas, responsible for 50% of the worldwide greenhouse effect?

43. As the Earth warms, do we get more or less of this gas in the atmosphere as a consequence? (Circle the correct answer).

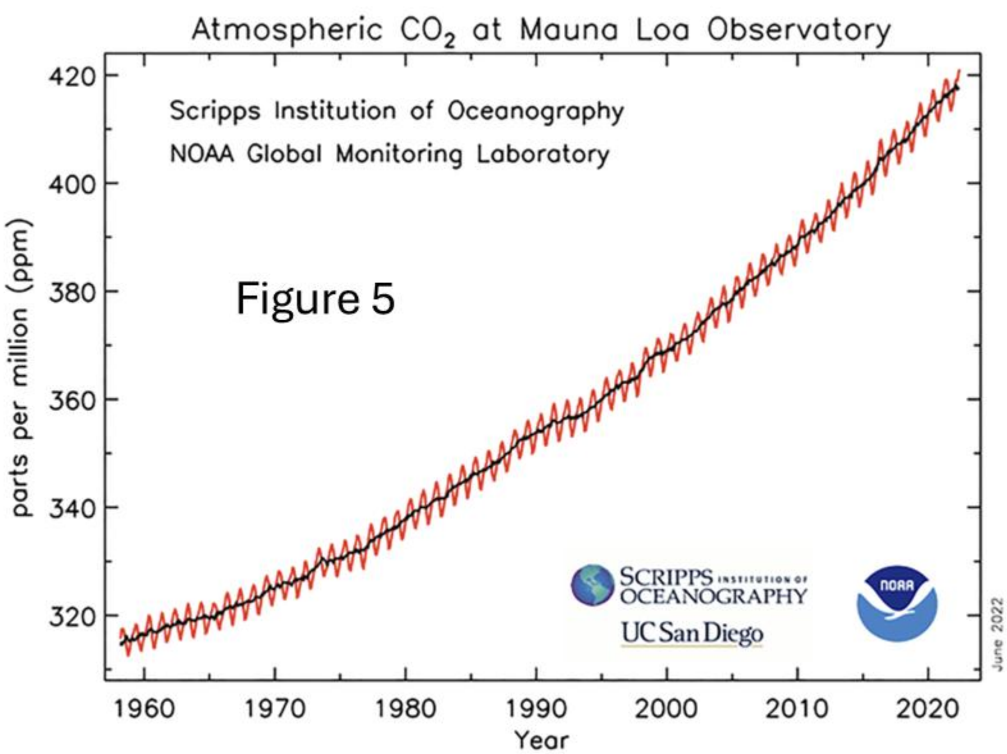
44. What is the name for the process described in Problem 43?

45. According to the EPA, which gas is considered the "most potent" greenhouse gas, 23,500 times more effective at trapping heat than CO_2?

46. Neglecting the Sun's natural variation, what is the name of the point in Earth's orbit when it receives the highest level of solar flux?

47. Approximately what date does this event occur?

Use Figure 5 to answer Problems 48-51:



48. (2 points) You've probably seen this graph a million times. Now let's see how much you've thought about and / or investigated the reasons for this graph. There is an oscillation of the CO_2 data around a central mean. What is the reason for the peaks and valleys?

49. Your answer to the previous problem should have included the words “Northern Hemisphere.” Why is this process tied to what goes on in the Northern Hemisphere?

50. Why is Mauna Loa a good place to monitor CO_2 concentrations?

51. What is the primary source of the increase shown in this chart?

52. Which process in the Nitrogen cycle involves bacteria known as diazotrophs?

53. Which process in the Nitrogen cycle involves organisms known as heterotrophs?

Section 4: Energy Balance Modeling and Climate Change (20 points)

54. (4 points) **NO CREDIT IF WORK IS NOT SHOWN**. The photosphere (visible surface of the Sun) has a temperature of roughly 5770 K. The radius of the Sun is 696,000 km. What is the luminosity of the Sun in watts?

55. (4 points) Mars has a radius of 3390 km, and is 228,000,000 km from the Sun. What is the "solar constant" for Mars (radiant flux density in W/m^2)? Use the accepted Sun luminosity of 3.828×10^{26} W, not the answer you got in the previous problem.

56. (4 points) Mars has an albedo of 0.25. What is the equilibrium temperature of Mars?

57. Earth's equilibrium temperature is 255 K. However, the actual mean surface temperature is 288 K. Why is this the case?

58. Mars has an atmosphere that is almost entirely (95%) carbon dioxide, yet the amount of warming is only about 3 K. Earth is roughly 33 K above equilibrium temperature. How is the warming effect so small when the amount of carbon dioxide is such a large percentage of the atmosphere?

59. Do volcanoes typically have a warming or cooling effect on the climate? (Circle the correct answer)

60. What chemical largely contributes to this effect?

61. (4 points) Rank these wavelength ranges in increasing order of their contribution to the amount of energy we receive from the Sun: x-ray, ultraviolet, visible, radio

1.
2.
3.
4.